



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	COMPUTATIONAL PHYSICS				
Type of Course	Core in Major				
Semester	III				
Academic Level	200-299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Basic computer knowledge.				
Course Summary	This course aims to equip students with computational and simulation methods in physics using Python programming. Numerical methods for differentiation, integration, solving differential equations, interpolation and curve fitting are introduced.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand computational thinking by learning Logical and algorithmic thinking.	U	F	Instructor-created exams.
CO2	Understand python syntax and write basic python programs using loops, several data types etc	U, Ap	F, P	Instructor-created exams / Practical Assignment

CO3	Understand Numpy and matplotlib modules and apply them to matrix manipulation and graphing data.	U, Ap	P	Instructor-created exams / Practical Assignment
CO4	Understand the significance of computational methods in physics.	U	F	Instructor-created exams / Seminar Presentation
CO5	Understanding the concepts of interpolation, curve fitting, numerical differentiation, integration and ODEs in physics using python	U, Ap	P	Instructor-created exams / Practical Assignment
CO6	Applying the computational and simulation methods to several branches of physics using python.	Ap	P	Instructor-created exams / Practical Assignment
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	THE COMPUTATIONAL THINKING		6	10
	1	Approach, Logical thinking, Algorithmic thinking	4	
	2	Writing algorithm - Sum of Two Numbers, Factorial Calculation, Bubble Sort, Fibonacci series up to a given term [More algorithms other than listed here can be given as assignments and can be asked for examination as application level questions].	2	
	Relevant Sections of Chapter 2 of Book 1			
II	THE PROGRAMMING LANGUAGE : PYTHON		19	20
	3	Print command, Data types in Python, Variables, Input statements, eval() and type casting, String operations.	1	
	4	Operators and Operator precedence, Expressions and Statements, Formatted printing,	2	
	5	List, Set, Tuple, Dictionary	2	
	6	Flow of Control : Sequential, Selective (simple if, if-else, nested	3	

		if, ladder if), Iterative (While, For), Continue, Break		
	7	File Input and Output, Pickling, User defined function. Built-in Functions.	2	
	8	Numpy : Arrays - creation, operations, eigenvalues solvers, dot, determinant, transpose, inverse, random number generation.	4	
	9	Matplotlib : Simple plot, Labelling axes, Title, Multiple plots, Subplots, Pie chart, Hist(), Polar plot, 3D plot - introduction	3	
	Relevant Sections of Book 2 and Book 3			
III	COMPUTATIONAL TECHNIQUES FOR EXPERIMENTAL PHYSICS		10	20
	10	Importance of Numerical Methods in experiments, Discretisation, Accuracy considerations.	1	
	11	Interpolation - Forward Difference Method - Newton's Formula for Interpolation.	2	
	12	Programs : Interpolation using experimental data*	1	
	13	Curve Fitting - Method of Least Squares : Linear, Linearization of Nonlinear Laws.	2	
	14	Programs : Curve fitting using experimental data*	1	
	15	Numerical Differentiation - 1st & 2nd order finite difference differentiation. Numerical Integration - Trapezoidal, Simpson's 1/3 Methods.	2	
	16	Root Finding Methods - Bisection, Newton-Raphson.	1	
	Sections : 3.1, 3.3.1, 3.6, 4.1, 4.2.1, 4.2.3, 6.2.3, 6.4.1, 6.4.2, 2.2, 2.5 of Book 4			
IV	COMPUTATIONAL TECHNIQUES FOR THEORETICAL PHYSICS		10	20
	17	Importance of Simulation in Physics. Solving First order ODE - Euler Method, Second Order Range-Kutta Method	2	
	18	Programs : Radioactive Decay*, Newton's Law of Cooling*	1	
	19	Solving 2nd Order ODE - Euler Method, Numerov's method	3	

	20	Programs : Configuration and Phase Space Plots of Simple and Damped Harmonic Oscillator*	2	
	21	Monte Carlo Method : Simple Integration - Hit or Miss Method, Mean-value Method (only)	1	
	22	Programs : Value of π *, Radioactive Decay*	1	
	Sections 8.4, 8.5 of Book 4 and 14.1, 14.2 of Book 5 [* Programs must be done using Python 3]			
V	PRACTICALS		30	
	Conduct any 6 experiments from the given list and 1 additional experiment, decided by the teacher-in-charge, related to the content of the course. The 7 th experiment may also be selected from the given list. Other experiments listed here may be used as demonstrations of the concepts taught in the course.			
	1	Solution of equations by bisection and Newton-Raphson methods - Implement the bisection method in Python from scratch. - Provide at least 4 functions with a specific mathematical equation and find the root using their implementation. - Analyze and explain the conditions under which the bisection method converges and discuss any potential pitfalls. - Similarly, implement the Newton-Raphson method in Python. - Provide the same or different functions and find the root using their implementation. - Compare the convergence speed of the Newton-Raphson method with the bisection method for different functions.		
	2	Least square fitting – straight line fitting - Write a code that fits a straight line to the data given and calculates the slope and intercept. - Plots the regression line along with the data points by giving, labels, title, legends and different colors - A real-world scenario or dataset can be used to apply linear regression to solve a practical problem.		
	3	Numerical Integration – Trapezoidal and Simpson's 1/3 rd rule - Implement the Trapezoidal and Simpson's 1/3 Rule in Python for a function given. - A physics scenario can be provided, where quantities like displacement, work, or energy are needed to calculate		

		through integration. Use both methods to perform the integration and interpret the results. Visualize the integration process by plotting the function and the areas under the curve corresponding to the Trapezoidal and Simpson's 1/3 Rule.		
	4	Simulation of projectile using Euler Method - Implement projectile motion simulation using the Euler method in Python. - Simulate the trajectory/ Plot using matplotlib (y vs x, y vs t and x vs t) - Compare with the theoretical values of range, maximum height and time of flight. - Change initial conditions such that the projectile is now a freely falling body. Plot y vs t. - Extend the simulation to include air resistance and compare the projectile motion with and without air resistance.		
	5	Simulation of simple and damped pendulums using RK2 Method - Simulates the damped pendulum and stores phase space coordinates to arrays using second order Runge-Kutta method. - Provide initial conditions and damping parameters for the damped pendulum scenario. - Plot the motion of the pendulum and phase space trajectories. - Change the Initial conditions and damping factor and analyse the results. Make sure turning the damping off reproduces the simple pendulum result.		
	6	Numerical differentiation using difference table. - Implement numerical differentiation using a difference table in Python. - Provide a function $y = f(x)$ and a set of data points. Compute the numerical derivative at specific points using the forward difference method. - Discuss the sensitivity of numerical differentiation to the choice of step size. - Present physics problems like compute the velocity or acceleration of a particle based on position data.		
	7	Monte- Carlo simulation of radioactive decay		

		● Implement a simulation of radioactive decay in Python. ● Provide initial conditions (number of particles, decay constant) and analyze the results, including plotting the decay curve over time. ● Calculate the half-life of the radioactive substance based on the simulation results and check how it compares to the theoretically expected half-life. ● Provide information about a specific radioactive isotope with a known half-life to simulate the decay of this isotope and compare the simulation results with the expected decay.		
8	Estimation of value of pi using Monte-Carlo Simulation	● Implement a Monte Carlo simulation to estimate the value of pi in Python. ● Analyze how the estimated value of pi converges as the number of samples increases. ● Create visualizations of the simulation results. Plot the points used in the simulation and visually demonstrate how the estimation of pi improves as more points are sampled.		
9	Solution system of linear equations and calculation of eigenvalues	● Solve a system of linear equations with three variables. ● Diagonalize a 3x3 matrix and verify that by evaluating the eigenvalues. Also evaluate the eigenvectors for the matrix. ● For better understanding, use Python (interactive mode) to verify that the eigenvector for an eigenvalue satisfies the eigenvalue equation: matrix times eigenvector equals eigenvalue times eigenvector.		
10	Least square fitting to an exponential function	● Take the data of transient effect in RC circuit (growth / decay) and write a code that fits an exponential function to the data and calculates the time constant. ● ExpEYES may be used to record the data. ● https://expeyes.in/experiments/electrical/rctransient.html ● https://expeyes.in/experiments/electrical/rltransient.html		
11	Taylor series- evaluation of sine and cosine	● Evaluate sine and cosine of a given angle, using Taylor expansion about zero. ● Print the difference with the built-in sine function. ● Analyse how the error reduces with the number of terms. ● Modify the program to calculate for higher angles to observe the effect of accuracy.		

Books and References:

1. Computational Thinking by Karl Beecher (Book 1)
2. A Student's Guide to Python for Physical Modeling by Jesse M. Kinder, Philip Nelson. Second Edition-Princeton University Press 2021 (Book 2)
3. Python for Education by Dr. B P Ajithkumar, IUAC, New Delhi; e-book freely downloadable from <https://scischool.in/python/index.html> (Book 3)
4. Introductory Methods of Numerical Analysis by S.S. Sastry, Fifth Edition (Book 4)
5. Basic Concepts in Computational Physics by Benjamin A. Stickler and Ewald Schachinger, Springer International Publishing Switzerland 2014 (Book 5)

Mapping of COs with PSOs and POs :

	PSO1	PSO 2	PSO 3	PSO 4	PSO5	PSO6	PO1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1	3	2	1	2	0	3	1	0	0	0	1	3	0
CO 2	0	3	2	0	0	3	0	3	0	0	0	3	0
CO 3	0	3	2	0	0	3	0	3	0	0	0	3	0
CO 4	0	2	3	0	0	3	0	3	0	0	0	3	0
CO 5	0	2	3	0	0	3	0	3	0	0	0	3	0
CO 6	0	3	3	0	0	3	0	2	1	2	3	0	0

Correlation Levels

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory / Practical Exam
- Assignments / Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory / Practical Exam	Assignment / Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	